TAL AVITAL SOFTWARE ENGINEER

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Ambitious Computer Science student with hands-on experience in software engineering. Passionate about embedded systems, specializes in embedded software, and thrives in a team environment.

Education

B.S. In Computer Science | University of Central Florida | Orlando, FL August 2024 – Present

Expected graduation: May 2027

- Studying programming techniques that prioritize established algorithms and data structures.
- Attending workshops, for GitHub, Linux, AI, Parallel Computing, Docker, and more.
- Attending events from tech companies to learn more about industry standards, expectations, and best practices.
- Experiencing the software development life cycle as both a developer and project lead.

Relevant Coursework | University of South Florida | Tampa, FL August 2022 – May 2024

- Took introductory courses: Calculus, Physics, and Statistics.
- Built a professional network through the Theta Tau Engineering Fraternity.

Technical Experience

Left is most experienced

System Architectures & Topologies: Bare-Metal C | RTOS (Task Scheduling) | Distributed Topologies (Client/Server, Central/Peripheral) | Concurrent Design | Communication Protocols (UART, I2C BLE, HTTP)

Software Tools: Git/GitHub | Embedded SDKs | Vim | Arch Linux | Linux Shell Scripting | Docker | Cloud Computing (AWS)

Hardware Lab (Familiar): Datasheet Analysis | PCB Design (EasyEDA) | Soldering & Prototyping | Multimeter Diagnostics

Professional Experience

Face Recognition Project, Part 2 | Personal Project June 2026 – Present

- Evolving the system hardware from a breadboard prototype to a self-designed and hand-soldered PCB.
- Integrating a battery-powered buck converter topology to establish voltage regulation and wireless deployment.
- Designing a weather-resistant enclosure to safeguard sensitive electronics from moisture and environmental elements.

Member Resume Database Project Lead | IEEE, UCF | Orlando, FL September 2025 – Present

- Leading the initiative for a front end that streamlines talent acquisition through IEEE UCF.
- Collaborating with IEEE UCF Outreach and Professional Development committees to map target audience requirements, define administrative data scopes, and align database fields with student profiles.
- Spearheading product discovery and user outreach by gathering feedback from students, university employers, and technical recruiters to continuously refine directory UI/UX.

Bluetooth Low-Energy Chat App | EEL 4932 - Internet of Things, UCF | Orlando, FL March 2026 – April 2026

- Engineered a Bluetooth Low Energy (BLE) wireless chat network using Silicon Labs' SDK.
- Implemented UART communication drivers to transfer data between a microcontroller and a host computer.
- Designed a terminal user interface to handle configuration states, device discovery, connection approvals, and user input.
- Utilized RSSI measurements to reliably trigger system alerts when signal degradation crosses critical thresholds.
- Routed low-level Bluetooth stack events that automate the program cycle.

Face Recognition Project, Part 1 | Personal Project December 2024 – January 2025

- Engineered a facial verification system to validate image captures against a whitelist database.
- Deployed an ESP32-CAM microcontroller to capture and offload raw JPEG binaries to an AWS Lambda HTTP endpoint.
- Architected an AWS cloud pipeline for computer vision processing, offloading intense AI algorithms from the edge device.
- Leveraged Docker to replicate the Amazon Linux environment on a Windows host to cross-compile the OS-specific dependencies.
- Link to GitHub repository: https://github.com/TAvital04/Facial-Recognition-Project.